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Novel secured speech communication for person authentication

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Abstract
Biometrics is the common method of securely and efficiently identifying and authenticating individuals by using unique biological features. Some common biometrics is fingerprint, speech, iris and signature. In this paper, the cryptosystem is proposed to enhance security and conserve the transmission bandwidth in implementing an authentication system The design of speech based secured authentication systems include the extraction of features from speech, creation of templates and testing procedures to authenticate the persons. The speaker recognition system is formed using Mel Frequency Cepstral coefficients (MFCC) and Recurrent Neural network (RNN) based machine learning technique. For developing a training system, MFCC features are extracted from the training data set. The RNN network is trained with features and a speakers' template is created for each speaker. In testing phase to ensure security in speech based authentication, MFCC features are extracted from the test speech set and these features are encrypted before it gets transmitted through the unsecured channel. The proposed crypto system is developed based on 3D logistic chaotic map and DNA operation. Firstly, MFCC features derived from the test speech set are concatenated and subjected to first level diffusion and confused using 3D logistic map. The resultant is encoded as a DNA sequence E(n), using any one of the eight rules for encoding DNA. The DNA XOR operation is performed between E(n) and 3D logistic map DNA sequence L(n). Finally, the encrypted feature set is attained by DNA decoding. In the test phase, the proposed system decrypts the features and is matched with a stored trained model to locate the identity of the speaker. Overall accuracy is 88% for the text independent and 96% for the text dependent person authentication system tested with genuine utterances. This research is extended to estimate the performance against attacks utterance and propose system is assessed with respect to rejection rate.

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1 Introduction

Biometric technologies are implemented by encoding a person's features as a codeword. Even if the people are twins, these characteristics are separate. Biometrics is biological measurements that are divided into two categories: invisible and visible biometrics. The attributes in invisible biometric systems are collected from the person's behaviour and retrieved indirectly; for example, signature and speech [27]. The features of visible biometric systems are taken directly from a person's physiological attributes, such as their fingerprints, retina, face, and iris. Extra hardware, such as an iris scanner, fingerprint sensor, and camera, is required for visible biometric systems. Speech is the most natural way for humans to communicate. For every person the confuration of vocal tracts, speaking styles, lungs, and vocal cords are dissimilar. Human speeches are recorded and utilized as biometrics. Speech-based biometric systems are less expensive than other biometrics.

Speaker recognition is a technique for identifying a speaker using speaking-specific knowledge. This speaker recognition system is classified as speaker identification and verification [24]. Speaker verification system compares an unknown speaker's voice to his or her template, and the speaker is recognized as genuine or imposter based on the one-to-one mapping. Speakers can be recognized by comparing the unknown speaker's voice to all-speaker templates and then identifying speakers via one on many mapping. Text independent and text dependent systems are two types of speaker identification and verification systems. In a text-independent system, different speech sets are utilised for training and testing, therefore memorising the utterances used for training is not necessary. In text-dependent systems, the same speech set is utilized for training and testing, and hence the speech sets are employed as a voice password.

A speech based authentication system enhances recognition rates by assuring that only the correct people have access to the correct data [33].Speech based systems are easily accessible and give better security compared with other biometric systems. The speech based authentication system is utilized for remote computer network access, mobile banking, forensics [4], and database access control. The design of speech based secured authentication systems include the extraction of features from speech, creation of templates and testing procedures to authenticate the persons. The features of the speech, which are unique to each person, are extracted using feature extraction techniques. The speech based person authentication system was developed using any one of the following feature extraction techniques such as Perceptual features, linear predictive cepstral coefficient (LPCC) and Mel frequency cepstral coefficient (MFCC). A significant number of feature vectors are formed from speech during the feature extraction stage. As a result, storing each feature vector takes up more memory [18]. The feature vectors are compressed and the data is deployed in such a way that the significant properties are kept using modelling approaches. Speaker recognition system uses following one modelling techniques Vector Quantization (VQ), Hidden Markov Model (HMM), Gaussian Mixture Model (GMM), Dynamic Time Warping (DTW), Support Vector Machine (SVM) and

Neural Networks. Many researchers have worked on speech-based person authentication systems over the years [1, 4, 5, 14, 18, 21, 22, 26, 28, 31, 33, 37]. Multilevel models such as Speaker verification module, text independent and voice password modules have been used to construct the person authentication system using voice [1]. The speech watermarking techniques [26] was created to protect speech biometrics and to improve the security of person authentication systems. Person authentication system was developed [22] using Perceptual, MFCC, LPCC features and VQ modelling is used to create templates. Person authentication system [21] is developed using MFCC features and K means clustering algorithm. The authentication systems [21, 22] performance are evaluated against genuine and spoofing attacks utterances. From literature survey, MFCC feature approximates the human system responses more precisely than any other features because the frequency bands are arranged logarithmically. Recurrent neural Network is one of the most popular machine learning algorithms and also outperforms other algorithms in both accuracy and speed. The proposed authentication system was developed based on MFCC features and RNN modeling technique. Proposed system performance was estimated with genuine and spoofing attacks utterances and compared with traditional systems.

The conventional method involves sending the speech via an unsecured channel to get access to the authentication mechanism. Intruders are especially vulnerable to this unsecured mode of communication, which could lead to attacks. Attackers will target the communication channel and allow access to biometric data, causing the authentication system to be hacked. Therefore, this system is unsafe. Such vulnerabilities result in a situation where biometric information needs to be protected in order to evade attackers. The primary plan of this research is to develop a cryptosystem to encrypt features and authenticate the persons with decrypted features. Chaotic sequences are periodic, divergent, and receptive to initial variables [14]. Many researchers have worked on developing speech encryption techniques based on chaotic maps [6, 9, 11, 12, 16, 18, 19, 23, 28–32]. The existing secured person authentication [21] was developed using multiple chaotic mapping techniques and Deoxyribonucleic acid (DNA) addition based speech crypto system. In this system [21], the test speech samples are encrypted and transmitted to the authentication system, which conserve more bandwidth. The objective of this research is to proposed cryptosystem based on the 3D logistic map sequences and DNA XOR operation to secure features and reduces bandwidth usage.

The proposed authentication system is implemented using MFCC features and recurrent neural network based machine learning techniques. Firstly, the system extracts features from each speaker's training speech set and creates templates for each speaker. After the separate speaker template is generated, the speaker will be able to access the authentication system from a remote location. The robust cryptosystem encrypts the test feature set to protect it from attackers. The decrypted feature set is compared to stored speaker templates, and a judgment is made as to whether a speaker can be identified or not. The authentication system efficiency is validated with decrypted and original feature sets.

The contributions of works are described as

I. Development of person authentication system based on MFCC features and recurrent neural network.

II. Development of 3D logistic and DNA encoding based cryptosystem to secure test features.
III. The performance of the authentication system is measured by recognition accuracy for both text-independent and text-dependent spoken utterances.
IV. Different metrics are used to evaluate the cryptosystem's efficiency, and the results show that the proposed encryption system is resistant to differential, statistical, and brute force attacks.

This paper is systemized as follows. Section 2 discusses the 3D logistic map and DNA encoding scheme. Section 3 highlights the Speech based person authentication system. Section 4 emphasizes the proposed cryptosystem. Section 5 discusses the performance of the proposed cryptosystem and authentication system. The conclusion of our research is revealed in Section 6.

2 Materials and techniques

2.1 AVSpoof database

The database comprises spoofing attacks that are reliable and impartial. The performances of the speaker identification system are influenced by the following major spoofing attacks: voice conversion, replay, speech synthesis, and impersonation [8]. This dataset can be used to test speaker recognition systems and anti-spoofing methods. The utterances of 31 male and 13female individuals are included in this database. Speech utterances are recorded at a sampling frequency of 44.1 kHz and then down sampled to 16 kHz. Several data gathering strategies, such as reading, passphrase and free speaking were committed by all speakers during each session. The free part and read speech part utterances are utilized for training in our work. Testing phase can be done by passphrase speech utterances. In our work, the male speaker's utterance is indicated as speaker1 to speaker31 and the female speaker's utterance is indicated as speaker32 to speaker44.

2.2 3D logistic map

Recently, chaos based cryptosystems have become attractive in terms of preserving speech content. Chaotic samples are defined as being periodic, divergent and sensitive to initial variables. The 3D logistic map sequence chaotic characteristics [17] are better than 1D logistic map sequences. It can be described as in (1)

$$\begin{aligned} x_{n+1} &= \alpha\, z_n^3 + \beta\, y_n^2 x_n + \gamma\,(1-x_n)x_n \\ y_{n+1} &= \alpha\, x_n^3 + \beta\, z_n^2 y_n + \gamma\,(1-y_n)y_n \\ z_{n+1} &= \alpha\, y_n^3 + \beta\, x_n^2 z_n + \gamma\,(1-z_n)z_n \end{aligned} \tag{1}$$

The value of γ is varying between 0.35 and 3.81. The value of β is taken between 0 and 0.022. α is varied between 0 and 0.015.

Table 1 DNA Mapping rules

	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6	Rule 7	Rule 8
A	00	00	11	11	10	01	10	01
T	11	11	00	00	01	10	01	10
G	01	10	01	10	11	11	00	00
C	10	01	10	01	00	00	11	11

2.3 DNA encoder

DNA encoding is an important method for medical, biological, forensic, diagnostic testing research and systemic medical areas [7]. In a DNA sequence, there are four distinct nucleic acids called guanine (G), Adenine (A), cytosine (C) and thymine (T).With respect to base pairing rules, Adenine usually matches with thymine and cytosine also matches with guanine. G and C can be considered complementary to one another, and similarly A and T. Table 1 display the DNA encoding rules. Table 2 indicates the rule 1 based DNA XOR operation.

3 Proposed person authentication system

The modules involved in the proposed system are depicted in Fig. 1. The authentication system includes training and testing phase. During training phase, speech utterances undergo pre-processing processes including VAD, pre emphasis, frame blocking and windowing to get windowed speech frames of 25 ms duration. The MFCC coefficients are extracted in each frame. RNN is used to train the neural network with features and develop trained models for all enrolled speakers. Once the trained model is created, the authentication system is accessed by the speaker from a remote location. In the testing phase, the test speech features are encrypted by the robust encryption system and encrypted features are transmitted to the authentication system to resist the channel attacks. The authentication system decrypts the encrypted features and decrypted feature set is coordinated with the template for each speaker, and a decision is made to identify a speaker. It is not necessary to extract features from speeches in the receiving system.

Table 2 DNA XOR Operation

⊕	**A**	**T**	**G**	**C**
A	A	T	G	C
T	T	A	C	G
G	G	C	T	A
C	C	G	A	T

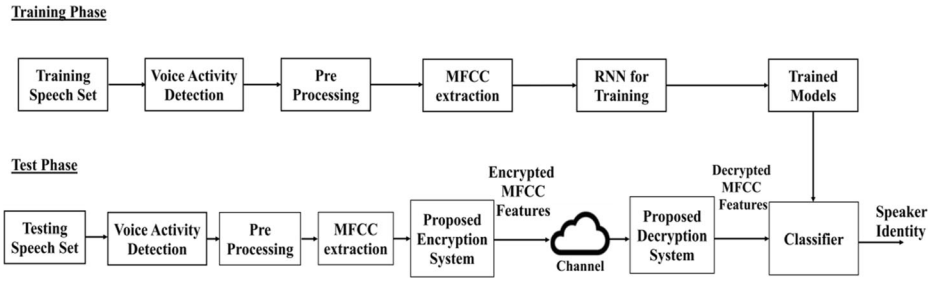


Fig. 1 Proposed Speech-based Authentication System

3.1 VAD

Improve the performance of the speech-based authentication system; VAD is employed to eliminate the silent section of the speech [36]. Zero crossing rate (ZCR) and short time energy (STE) are used for VAD. The outputs of the vocal tract system, which are stimulated by the vibrations of the vocal cords, are referred to as speech. The vocal cords' vibrations produce quasi-periodic pulses that stimulate the vocal tract for voiced sounds and noise like stimulation for silence. Characterizing this time-varying system is quite difficult. So, speech signal is divided into short length frames (10 to 30 milliseconds) and the vocal tract system is modified to be time invariant. The number of times the zero crossing occurs in each frame is referred to as the ZCR. The value of ZCR should be less for silence part of the speech segment. A short time ZCR is a weighted average of the number of times the signal changes sign inside a frame. It is expressed as in (2)

$$zcr(k) = \sum_{n=-\infty}^{\infty} |sgn(d(n)) - sgn(d(n-1))| \; w(n-k) \tag{2}$$

Signum $d(n)$ is expressed in eq. (3)

$$sgn(d(n)) = \begin{Bmatrix} 1 & if\ d(n) \geq 0 \\ -1 & if\ d(n) < 0 \end{Bmatrix} \tag{3}$$

Each speech segment's STE is derived by summing the sample squares. The energy of the voice section is greater than the energy of the silence section. STE can be described as in (4)

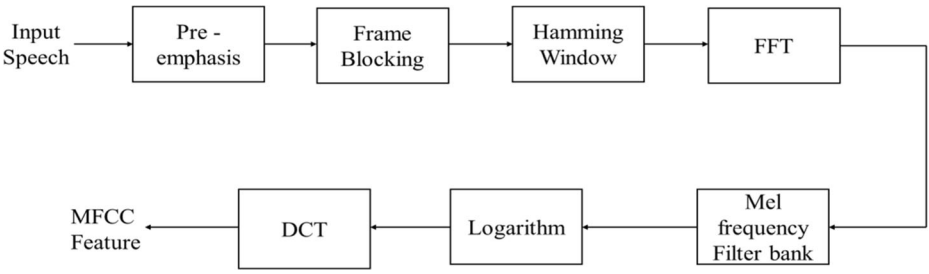


Fig. 2 MFCC extraction

$$E(k) = \sum_{n=-\infty}^{\infty} d^2(n) w(n-k) \tag{4}$$

$d(n)$- Speech Samples
$w(n)$- Hamming Window

3.2 MFCC estimation

The features of the speech, which are unique to each speaker, are extracted using feature extraction techniques. Speech signals produced by the vocal tract can be analysed more easily in the frequency domain than in the time domain. Figure 2 illustrates the MFCC estimation block diagram. In the pre emphasis step, the high pass filter is used to spectrally flatten the voice signals. Speech is typically time varying signals, thus they are separated into short segments (typically 10-30 ms) for assimilating speech signals as output of the time-invariant vocal tract system. Hamming window sequences are used to remove the discontinuity at the start and end of each speech segment. The spectrum of each frame is calculated using Fast Fourier transform (FFT) and described as in (5)

$$Y_n(\omega) = FFT(a(n) * w(n)) = A_n(\omega) * W(\omega) \tag{5}$$

Frequency response of Mel filter banks is shown in Fig. 3. The spectrum of Mel scale filter bank is multiplied with $Y_n(\omega)$ and the resultant is transformed into the Mel scale, represented as in (6). The Mel scale is an auditory scale that simulates the sensitivity of a human ear to different sounds.

$$Z_n(\omega) = Y_n(\omega) * T(\omega) \tag{6}$$

Mel frequency cepstral coefficient is the result of discrete cosine transformation applied to the logarithm of the filtered signal. It can be expressed as in (7)

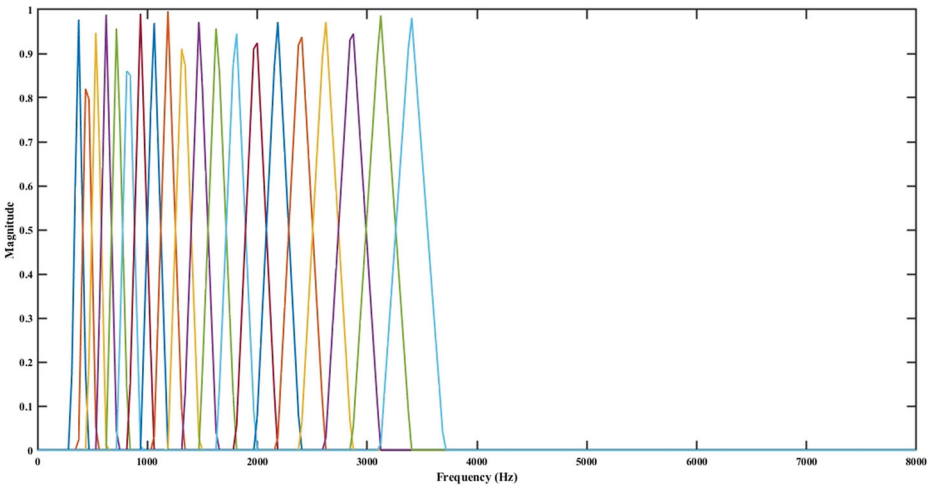


Fig. 3 Frequency Response of Mel Filter bank

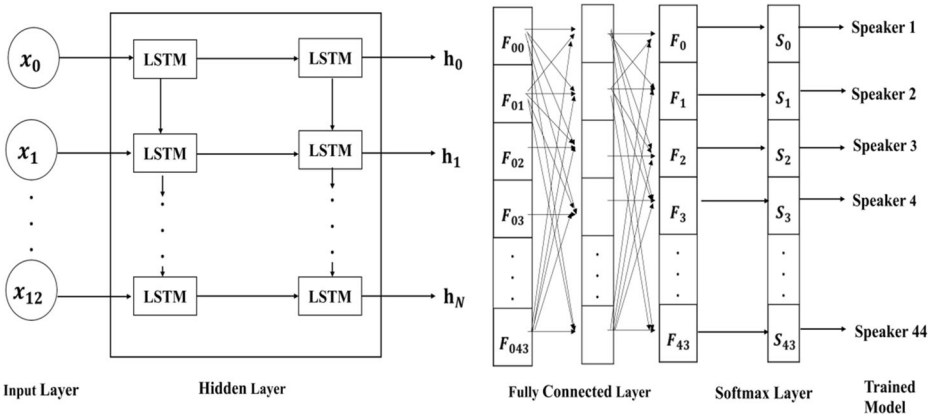


Fig. 4 Architecture of RNN

$$M_n(k) = \sum_{m=1}^{N} \log(|Z_n(m)|) \cos\left(k(m-0.5)\frac{\Pi}{N}\right), \quad k = 1, 2, \ldots..N \tag{7}$$

3.3 RNN

A network with more than one hidden unit is commonly called a deep learning neural network. RNNs are frequently used to detect context-based pattern tasks like speech recognition. During the RNN process, each neuron holds a memory cell. Long short term memory (LSTM) [10] is an RNN architecture that was created to resolve long-term dependencies. RNN's architecture is depicted in Fig. 4.

3.3.1 LSTM layer

A hidden layer, also known as an LSTM, is made up of a set of interconnected blocks known as memory blocks. The construction of the LSTM layer is shown in Fig. 5. Figure 6 depicts the internal workings of the hidden node. The states forget gate, input gate, update, and output gate are all contained in the hidden layer.

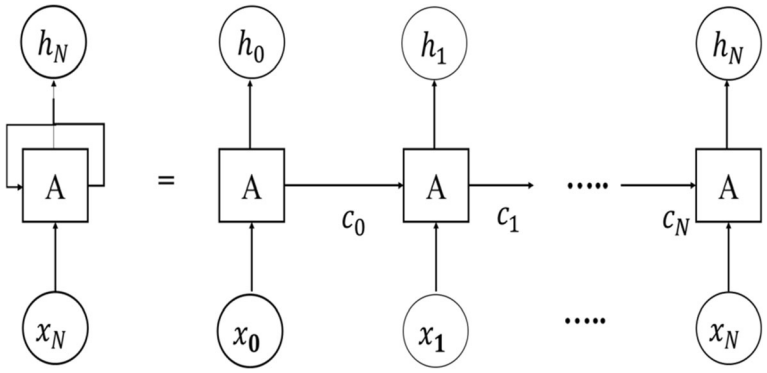


Fig. 5 LSTM Layer Structure

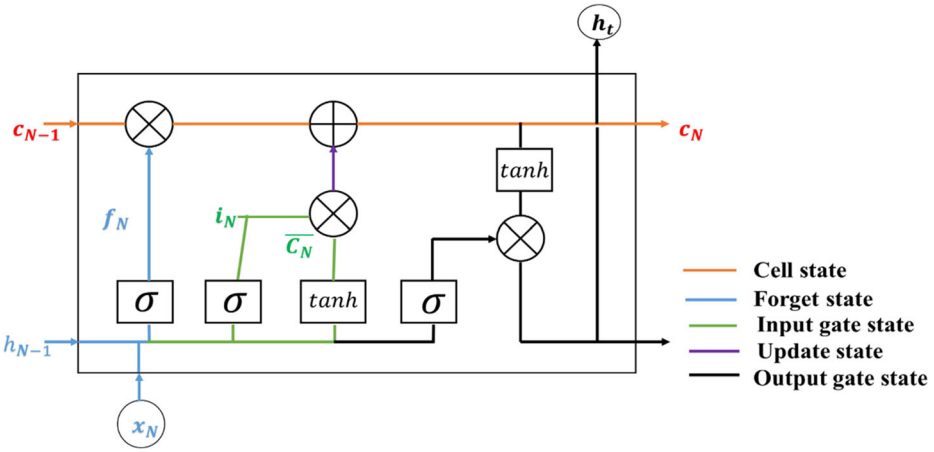


Fig. 6 Hidden layer Neutron internal operation

LSTM depends on the cell state. The LSTM has the ability to add and remove data from the cell state. The initial step in the LSTM process is to figure out what data should be sent to the cell state. The sigmoid layer makes this information and named forget gate state and represented as in (8)

$$f_N = \sigma\left(w_f.[x_N, h_{N-1}] + b_f\right) \tag{8}$$

The LSTM's second stage involves determining which data can be stored in the cell state. It is achieved in two stages. To begin, we'll use a sigmoid layer called input gate state to establish what values we'll be upgrading to. Then, using a tanh layer, a variable is created that may be added to the state with new values. In the next step, combine these two sections to create an update to the state. These two sections are written as in (9) and (10)

$$i_N = \sigma(w_i.[x_N, h_{N-1}] + b_i) \tag{9}$$

$$\overline{C_N} = tanh\ (w_c.[x_N, h_{N-1}] + b_c) \tag{10}$$

The new cell state C_N is achieved from old cell state C_{N-1}, and it is attained by multiplying C_{N-1} with f_N and resultant is added with $i_N * \overline{C_N}$. It is represented as in (11)

$$C_N = f_N * C_{N-1} + i_N * \overline{C_N} \tag{11}$$

The hidden layer output depends on cell state. First, run a sigmoid layer to figure out which part of the cell state we can get as an output. Then bring C_N through tanh and multiply it by sigmoid gate output, then actual output of the network is obtained. These are represented as in (12), and (13)

$$O_N = \sigma(w_o.[x_N, h_{N-1}] + b_o) \tag{12}$$

$$h_N = O_N * \tanh(C_N) \tag{13}$$

3.3.2 Fully connected layer

The final LSTM data is flattened and then put into the fully connected layers. Neurons, weights, and biases are all included in this unit. All input neurons in the fully linked layer are connected to each activation function of the next unit neuron. Each neuron performs the computation as in (14)

$$F_i = \sigma(P_i * W + b) \tag{14}$$

b- Bias vector
W- Weight matrix
P_i- All input from previous neuron unit

3.3.3 Softmax layer

Softmax is the final layer in the LSTM network. It can be found at the last part of fully connected network. This layer employs the softmax activation function after passing through the fully connected layer. For multiclass classification, the probability of each class can be computed. It's written as in (15)

$$S_i = \frac{\exp(F_i)}{\sum_j F_j} \tag{15}$$

4 Proposed crypto system to secure feature sets

The proposed encryption system is developed based on 3D logistic mapping technique and DNA computing for encrypting the MFCC features. Speech is usually divided into short duration speech frames. Thirteen MFCC coefficients are extracted from each speech frame and concatenated to create a single feature set, which is provided as input to the encryption system.

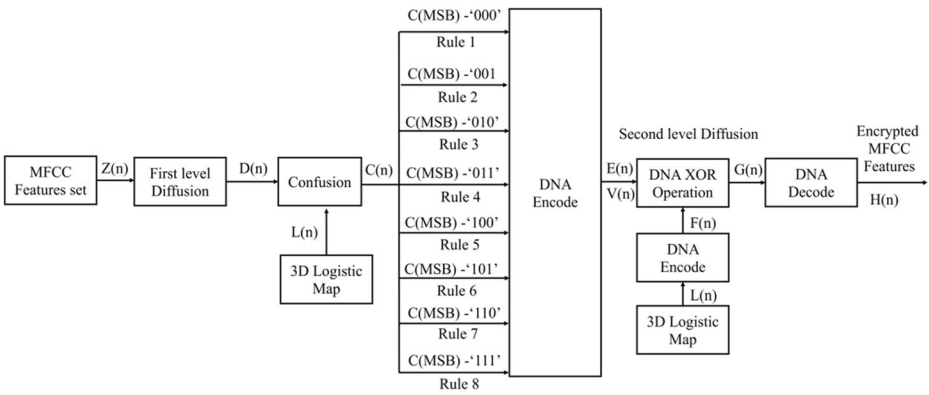


Fig. 7 Proposed Encryption System

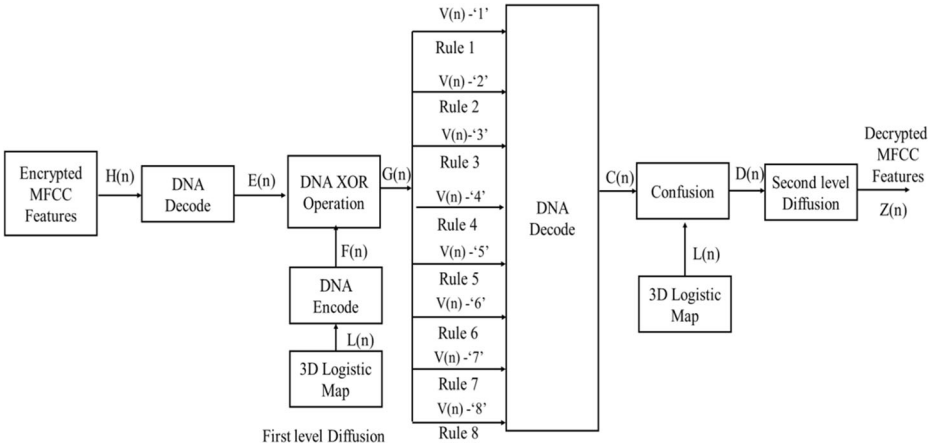


Fig. 8 Proposed Decryption System

Figures 7 and 8 describe the block diagram of the proposed encryption system and decryption system. The encryption process is configured to contain four phases: key generation, first level diffusion phase, confusion phase, and second level diffusion phase.

4.1 Key generation using 3D logistic map and secured hash Algorithm 3(SHA 3)

The 3D logistic map requires three initial conditions (x_0, y_0, z_0) and three control parameters (α, β, γ) to generate pseudo random numbers, which is obtained from hash values. The 256 bits hash key is generated from input future set using SHA 3 algorithm. This 256 bit key is separated into 8 bit segments. It is represented as in (16)

$$k = \{k_1, k_2, \ldots..k_{32}\} \tag{16}$$

The initial conditions x_0, y_0, z_0 are obtained using hash value k.

$$x_0 = \frac{\left((k_1 \oplus k_2 \oplus k_3) + \sum_{n=4}^{6} k_n\right)}{2^{10}} \tag{17}$$

$$y_0 = \frac{\left((k_7 \oplus k_8 \oplus k_9) + \sum_{n=10}^{12} k_n\right)}{2^{10}} \tag{18}$$

$$z_0 = \frac{\left((k_{13} \oplus k_{14} \oplus k_{15}) + \sum_{n=16}^{18} k_n\right)}{2^{10}} \tag{19}$$

The control parameters α, β, γ are obtained based on algorithm 1.

Algorithm 1 -Pseudo code for generating control parameters of 3D logistic Map

Input: $k_{19}, k_{20}, \ldots\ldots, k_{32}$

Output: α, β, γ

Initialize: $k = 0$

FOR $i \leftarrow 19$ *to* 32

$k \leftarrow k \oplus k_i$

END FOR

$k_1 \leftarrow k$

IF $0 < k_1 < 0.015$

$\alpha \leftarrow k_1$

ELSE

$\alpha \leftarrow k_1 \; mod \; 0.014$

END IF

$k_2 \leftarrow k$

IF $0 < k_2 < 0.022$

$\beta \leftarrow k_2$

ELSE

$\beta \leftarrow k_2 \; mod \; 0.021$

END IF

$k_3 \leftarrow k$

IF $0.35 < k_3 < 3.81$

$\gamma \leftarrow k_3$

ELSE

$k_4 \leftarrow k_3 \; mod \; 3.4$

$k_5 \leftarrow k_4 + 0.36$

$\gamma \leftarrow k_5$

END IF

4.2 First level diffusion phase

In the first level diffusion phase, the value of each feature is changed without changing their place in the block. Figure 9 shows the operation of the first level diffusion phase. Each feature is subjected to two stage bit level XOR and XNOR operation. The original feature set is represented as $Z(n) = \{ Z_1(n), \; Z_2(n), \; .., \; Z_N(n) \}$. N – Total features contained in the feature set.

The first stage bit level XOR and XNOR operation of $Z_2(n)$ feature is performed between previous feature $Z_1(n)$ and next feature $Z_3(n)$. It is represented in (20) and (21)

$$I_i = XOR(Z_{i-1}, Z_i) \tag{20}$$

$$K_i = XNOR(I_i, Z_{i+1}) \tag{21}$$

The second stage bit level XOR and XNOR operation of $K_2(n)$ feature is performed between next features $K_3(n)$ and $K_4(n)$. It is represented in (22) and (23)

$$J_i = \mathrm{X}OR(K_i, K_{i+1}) \tag{22}$$

$$D_i = XNOR(J_i, K_{i+2}) \tag{23}$$

Algorithm 2 - Pseudo code for First level diffusion phase

Input: $\{Z_1(n),\ Z_2(n), \ldots., Z_N(n)\}$
Output: $\{D_1(n),\ D_2(n), \ldots., D_N(n)\}$
FOR $i \leftarrow 1\ to\ N$
$k \leftarrow i-1$
IF $k = 0$
$k \leftarrow N$
END IF
$I_i(n) \leftarrow Z_k \oplus Z_i$
$j \leftarrow i$
IF $j = N$
$j \leftarrow 0$
END IF
$K_i(n) \leftarrow I_i \odot Z_{j+1}$
END FOR
FOR$i \leftarrow 1\ to\ N$
IF$i < N$
$J_i(n) \leftarrow K_i \oplus K_{i+1}$
ELSE
$J_i(n) \leftarrow K_i \oplus K_1$
END IF
$k \leftarrow i$
IF$k < N-1$
$D_i(n) \leftarrow J_i \odot K_{k+2}$
ELSE IF $j = N-1$
$D_i(n) \leftarrow J_i \odot K_1$
ELSE IF $j = N$
$D_i(n) \leftarrow J_i \odot K_2$
END IF
END FOR

4.3 Confusion phase

In this phase, the position of the $D(n)$ is changed with respect to key $L(n)$, without altering their value. This phase creates a high correlation between neighbouring values by changing the entire samples into a new position. The 3D logistic map sequences $L(n)$ are organized in

ascending order and expressed as $l_1(n)$. The index of $D(n)$ is arranged based on the index of $l_1(n)$ and expressed as in (24)

$$C(n) = D(index(l_1)) \tag{24}$$

4.4 Second level diffusion phase

In this phase, the DNA XOR is achieved among the DNA sequence of confused output $C(n)$ and DNA sequence $L(n)$. From eight DNA encoding rules, the selection of encoding rule to encode $C(n)$ depends on the first three MSB bits of their value. The DNA rule of each feature is stored in $V(n)$. The logistic map sequences are encoded into $F(n)$ using DNA coding rule 1. It is expressed as in (25)

$$G(n) = E(n) \oplus F(n) \tag{25}$$

The binary sequence $g(n)$ is recovered and transformed to decimal form after applying the DNA decoding rule 1 to $G(n)$. This is the encrypted MFCC features $H(n)$.

Algorithm 3 - Pseudo code for Second level diffusion phase

Input: $\{L_1(n),\ L_2(n), \ldots, L_N(n)\}$, $\{C_1(n),\ C_2(n), \ldots, C_N(n)\}$,

Output: $\{G_1(n),\ G_2(n), \ldots, G_N(n)\}$

FOR $i \leftarrow 1\ to\ N$

IF $C_i('MSB') = '000'$

$E_i(n) \leftarrow Rule\ 1\ DNA\ encode\ of\ C_i(n)$

ELSE IF $C_i('MSB') = '001'$

$E_i(n) \leftarrow Rule\ 2\ DNA\ encode\ of\ C_i(n)$

ELSE IF $C_i('MSB') = '010'$

$E_i(n) \leftarrow Rule\ 3\ DNA\ encode\ of\ C_i(n)$

ELSE IF $C_i('MSB') = '011'$

$E_i(n) \leftarrow Rule\ 4\ DNA\ encode\ of\ C_i(n)$

ELSE IF $C_i('MSB') = '100'$

$E_i(n) \leftarrow Rule\ 5\ DNA\ encode\ of\ C_i(n)$

ELSE IF $C_i('MSB') = '101'$

$E_i(n) \leftarrow Rule\ 6\ DNA\ encode\ of\ C_i(n)$

ELSE IF $C_i('MSB') = '110'$

$E_i(n) \leftarrow Rule\ 7\ DNA\ encode\ of\ C_i(n)$

ELSE IF $C_i('MSB') = '111'$

$E_i(n) \leftarrow Rule\ 8\ DNA\ encode\ of\ C_i(n)$

END IF

$F_i(n) \leftarrow Rule\ 1\ DNA\ encode\ of\ L_i(n)$

$G_i(n) \leftarrow E_i(n) \oplus F_i(n)$

END FOR

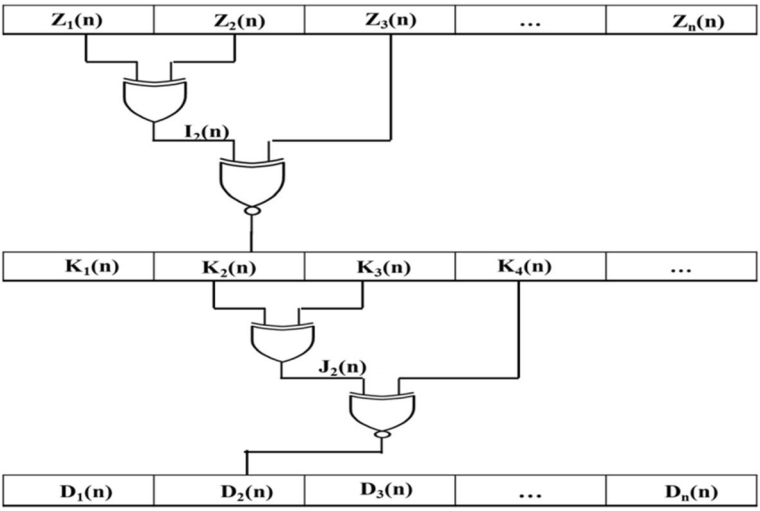


Fig. 9 First level Diffusion phase

4.5 Proposed decryption system

The operations of the decryption system are the inverse of the encryption system's operations. The encrypted MFCC features $H(n)$ and chaotic map sequence $L(n)$ are encoded to the DNA sequence using rule 1 DNA mapping. The DNA XOR is achieved among $L(n)$ and $H(n)$. It is expressed as in (26)

$$G(n) = E(n) \oplus F(n) \tag{26}$$

From eight DNA decoding rules, the selection of decoding rule to decode $G(n)$ depends on the stored DNA rules in V(n). The decoded binary sequences are confused with 3D logistic map sequences $L(n)$ and resultant $D(n)$ is subjected to second level diffusion. In the second level diffusion phase, the first stage bit level XOR and XNOR operation of $D_2(n)$ feature is performed between previous feature $D_1(n)$ and next feature $D_3(n)$. It is represented as in (27) and (28)

$$X_i = XOR(D_{i-1}, D_i) \tag{27}$$

$$K_i = XNOR(X_i, D_{i+1}) \tag{28}$$

The second stage bit level XOR and XNOR operation of $K_2(n)$ feature is performed between next features $K_3(n)$ and $K_4(n)$. It is represented as in (22) and (23)

$$Y_i = XOR(K_i, K_{i+1}) \tag{29}$$

$$Z_i = XNOR(Y_i, K_{i+2}) \tag{30}$$

The resultant $Z(n)$ of second level diffusion is the decrypted feature set.

5 Results and discussion

The initial conditions (x_0, y_0, z_0) and control parameters (α, β, γ) of 3D logistic map serve as a secret key. These keys are obtained from hash values. The total neurons present in the input layer are same as the number of features extracted in each speech segment, in this work the input layer size is 13. Set the hidden or LSTM layer with an output size of 100. The size of softmax layer and fully connected layer is 44, which is same as the number of speakers to be authenticated. From the simulated results of the proposed cryptosystem, it is revealed that the encrypted features as in Fig. 10b is entirely dissimilar from the original features as in Fig.10a. The decrypted features Fig. 10c is the similar as the original features.

5.1 Security analysis test

Efficiency of the crypto system was estimated to check whether it combats the statistical, brute force and differential attacks. The NFCR test and UACI test are used to investigate the encryption system resist differential attack. Correlation coefficient, PSNR, SNR and histogram are the statistical parameters considered to investigate the encryption system resist statistical attack.

5.1.1 UACI and NFCR analysis

To find the proper key or original features, the attacker compares variations in input characteristics to changes in encrypted features. This differential attacks defeats the proposed cryptosystem. To prevent this attack, the cryptosystem must be sensitive to minor alterations in input features and key elements. Number of Features Change Rate (NFCR) and Unified Average changing Intensity (UACI) are used to analyse the encryption system against differential attack [35]. Let $z(n)$ be the original feature set. The encryption of $z(n)$ is represented as EF_1. Let $P(n)$ is the one feature change of $z(n)$. $P(n) = z(n)/_{z(0) = z(0) + 0.5}$. Same set of keys are used to encrypt $P(n)$ and expressed as EF_2. The NFCR is used to calculate the proportion of distinct features in the two encrypted feature sets EF_1, EF_2. NFCR is represented as in eq. (31)

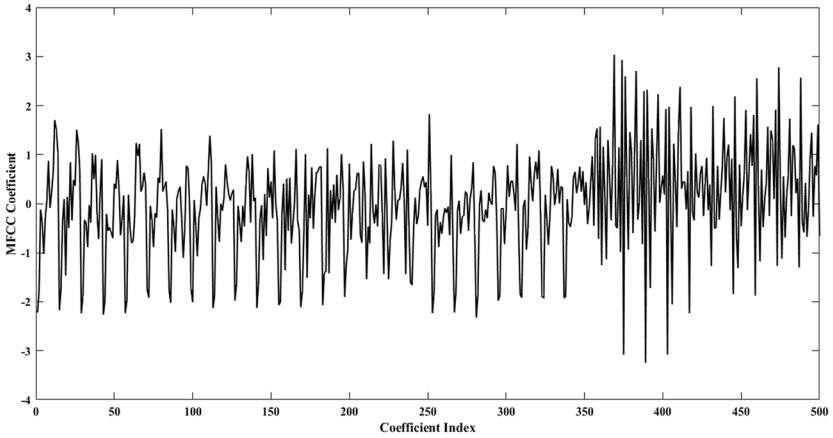
$$NFCR = \frac{\sum_i w(i)}{N_F} \times 100 \tag{31}$$

$$w(i) = \begin{cases} 0 & if\ EF_1(i) = EF_2(i) \\ 1 & if\ EF_1(i) \neq EF_2(i) \end{cases} \tag{32}$$

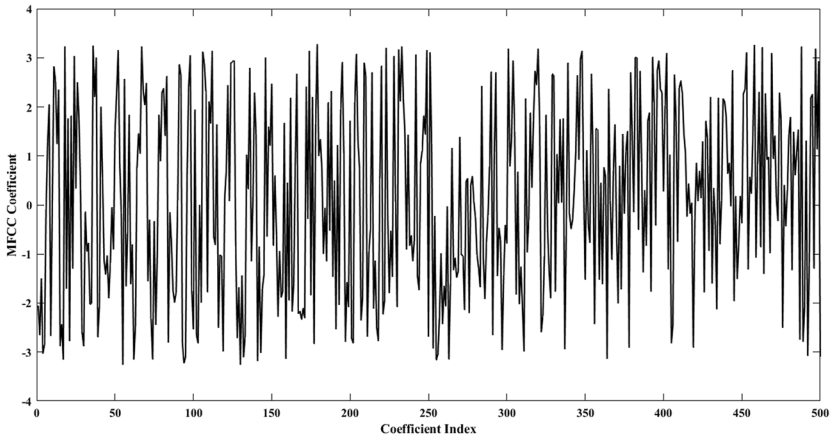
The average intensity of the two encrypted features' changes is calculated using the UACI. It is decribedas in eq. (33)

$$UACI = \frac{1}{N_F}\left[\sum_i \left|\frac{EF_1(i) - EF_2(i)}{65535}\right|\right] \times 100\% \tag{33}$$

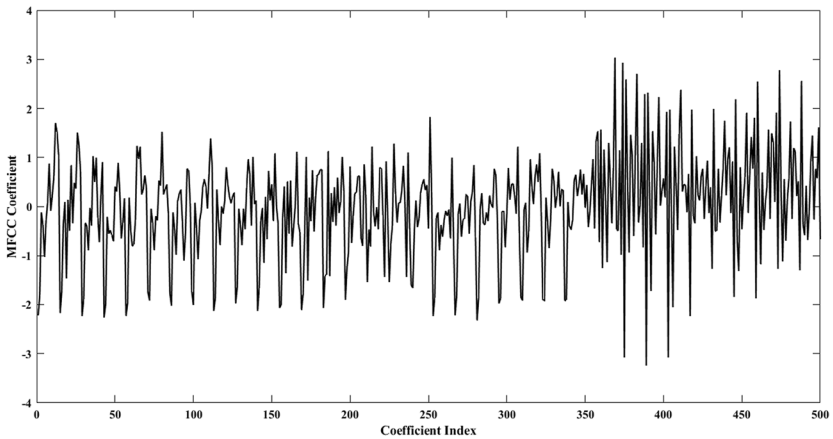
N_F= Total Number of the encrypted features
EF_1= Encrypted features obtained from $z(n)$
EF_2= Encrypted features obtained from $P(n)$



(a)



(b)



(c)

Fig. 10 **a** Original MFCC Feature set of one speaker. **b** Encrypted MFCC Feature set. **c** Decrypted MFCC Feature set

The ideal NFCR and UACI values must be greater than 99% and 33%. The NFCR and UACI values for the 44 different speaker feature sets are included in Table 3.These findings suggest that the proposed feature encryption system's NFCR and UACI values are much closer to optimum values. As a result, there is a lot of difference between the encrypted features EF_1, EF_2 and the proposed encryption method can withstand differential attacks.

Table 3 Text dependent Authentication System – Security Analysis

SI. No	Speaker	NSCR	UACI	SNR among Encrypted and Original Features	PSNR among Encrypted and Original Features	Correlation Coefficient among Encrypted and Original Features
1	Speaker 1	99.9977	36.5600	−91.5561	−91.5554	0.0027
2	Speaker 2	99.9978	39.6590	−91.5606	−91.560	0.0017
3	Speaker 3	99.9980	39.5816	−91.5479	−91.5473	0.0015
4	Speaker 4	99.9982	39.1032	−91.552	−91.5515	0.0025
5	Speaker 5	99.9983	39.7018	−91.5556	−91.5551	0.0016
6	Speaker 6	99.9981	36.4175	−91.5501	−91.5495	−0.0031
7	Speaker 7	99.9981	39.6492	−91.5615	−91.5609	−0.0018
8	Speaker 8	99.9977	38.5117	−91.5618	−91.5611	−0.0034
9	Speaker 9	99.9978	37.4404	−91.5611	−91.5604	0.0026
10	Speaker 10	99.9978	39.1005	−91.5647	−91.564	0.0026
11	Speaker 11	99.9978	38.5899	−91.5559	−91.5552	−0.0018
12	Speaker 12	99.9980	36.6808	−91.5458	−91.5451	−0.0067
13	Speaker 13	99.9975	38.1698	−91.5527	−91.5519	0.00085
14	Speaker 14	99.9984	39.1099	−91.5535	−91.553	0.00222
15	Speaker 15	99.9979	39.1285	−91.5618	−91.5612	0.00032
16	Speaker 16	99.9981	38.6038	−91.5379	−91.5373	−0.00069
17	Speaker 17	99.9977	36.4052	−91.5574	−91.5567	−0.00074
18	Speaker 18	99.9978	37.6127	−91.5553	−91.5547	0.00086
19	Speaker 19	99.9974	39.1078	−91.5612	−91.5604	0.00097
20	Speaker 20	99.9979	37.9516	−91.5676	−91.567	−0.0014
21	Speaker 21	99.9976	39.7133	−91.5493	−91.5486	−0.0035
22	Speaker 22	99.9978	39.1282	−91.5653	−91.5646	−0.00033
23	Speaker 23	99.9978	36.5844	−91.541	−91.5403	−0.0014
24	Speaker 24	99.9976	36.4170	−91.5792	−91.5785	−0.0043
25	Speaker 25	99.9976	38.2259	−91.5871	−91.5864	−0.0075
26	Speaker 26	99.9980	36.4284	−91.5743	−91.5737	−0.0018
27	Speaker 27	99.9982	39.6630	−91.5698	−91.5693	−0.0038
28	Speaker 28	99.9979	38.9695	−91.5606	−91.5599	0.00099
29	Speaker 29	99.9979	36.7340	−91.569	−91.5683	−0.00133
30	Speaker 30	99.9979	36.6703	−91.5564	−91.5557	−0.0021
31	Speaker 31	99.9979	38.2783	−91.5713	−91.5707	−0.0026
32	Speaker 32	99.9978	37.1091	−91.561	−91.5603	0.00051
33	Speaker 33	99.9981	34.1229	−91.5676	−91.567	0.00475
34	Speaker 34	99.9980	39.5174	−91.5614	−91.5608	−0.00181
35	Speaker 35	99.9979	39.4698	−91.5567	−91.5561	0.00347
36	Speaker 36	99.9981	36.4323	−91.5574	−91.5568	−0.00190
37	Speaker 37	99.9981	39.5757	−91.5764	−91.5758	−0.00168
38	Speaker 38	99.9978	39.1738	−91.5565	−91.5558	0.00096
39	Speaker 39	99.9981	39.1038	−91.5583	−91.5578	−0.00061
40	Speaker 40	99.9977	38.4734	−91.5916	−91.5909	−0.00287
41	Speaker 41	99.9982	39.1742	−91.560	−91.5595	−0.00357
42	Speaker 42	99.9979	36.1184	−91.5810	−91.5804	−0.00087
43	Speaker 43	99.9981	35.9593	−91.5539	−91.5533	0.00036
44	Speaker 44	99.9974	38.9402	−91.5668	−91.566	0.00065

5.1.2 SNR test

The SNR is a commonly used method for calculating the level of noise in the encrypted features. If the value of SNR is negative, it indicates that the signal power is lower than the noise power, making the encrypted features undetectable. The SNR can be expressed as in (34)

$$SNR\ (dB) = 10\ log_{10} \frac{\sum_{j=1}^{N_F} z^2(j)}{\sum_{j=1}^{N_F} [z(j) - h(j)]^2} \tag{34}$$

Where $h(j)$ is the encrypted feature set; SNR test results for the feature set of 44 different speakers are included in Table 3. The SNR between the original and encrypted feature sets is negative in our proposed encryption technique. It shows that the proposed crypto system is work well and encrypted features are highly undetectable.

5.1.3 PSNR test

PSNR [20] is the ratio of the original feature set's peak power to the mean square error (MSE) between the original and encrypted feature sets and expressed as in (35)

$$PSNR = 10\ log_{10} \frac{N_F z^2}{\|z - h\|^2} \tag{35}$$

where 'z' denotes the peak power of the original feature set and 'N_F' denotes the length of feature set. $\|z - h\|^2$ denotes MSE. Table 3 includes the results of the PSNR test for the feature set of 44 different speakers. The PSNR values between the original and encrypted feature sets are negative, implying that the proposed encryption method can withstand statistical attacks.

5.1.4 Correlation test

Correlation between original and encrypted features rc_{zh} is the covariance of the original and encrypted features normalized by the variances of original and encrypted features. Correlation test can be used to determine the correlation between encrypted and original features. The correlation plot of the original feature set and encrypted feature set is shown in Figs. 11 and 12. From Fig.11, the strong correlation between the features in the original feature set and Fig. 12 depicts the low correlation between features in the encrypted feature set. This indicates that the proposed encryption system is good. The correlation coefficient can be expressed as in (36)

$$rc_{zh} = \frac{\frac{\sum_{j=1}^{N_F} (z(j) - E(z))\ \ (h(j) - E(h))}{N_F}}{\sqrt{\frac{\sum_{j=1}^{N_F} (z(j) - E(z))^2}{N_F}}\ \ \sqrt{\frac{\sum_{j=1}^{N_F} (h(j) - E(h))^2}{N_F}}} \tag{36}$$

N_F is the number of features presented in the original feature set. $E(z)$, and $E(h)$ indicate the mean of original feature set z and encrypted feature set h. It is computed as in (37) and (38)

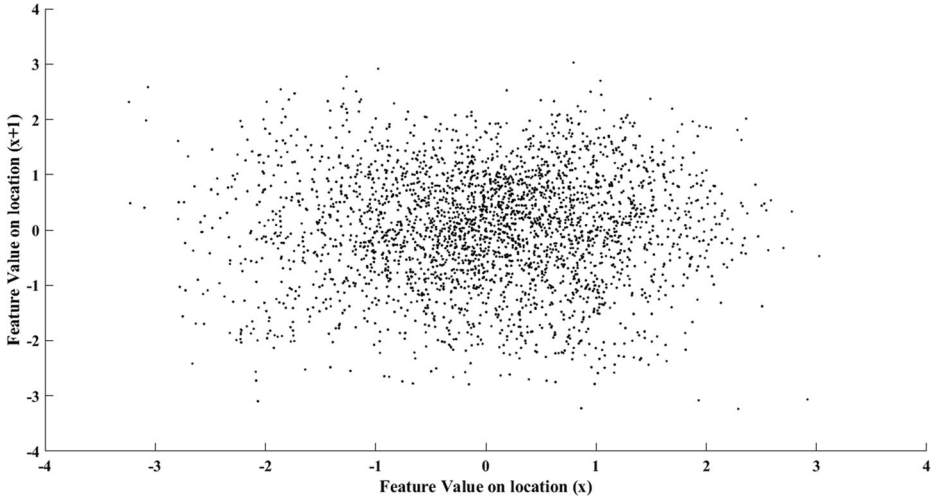


Fig. 11 Correlation plot of Original Feature set

$$E(z) = \frac{1}{N_F} \sum_{j=1}^{N_F} z(j) \tag{37}$$

$$E(h) = \frac{1}{N_F} \sum_{j=1}^{N_F} h(j) \tag{38}$$

Table 3 includes results of the correlation coefficient test for the feature set of 44 different speakers. According to the results, the correlation coefficient value is nearly zero, indicating that the encrypted feature set is random and uncorrelated with the original feature set. As a result, the original and encrypted features have very different values, and the proposed encryption scheme can withstand a statistical assault.

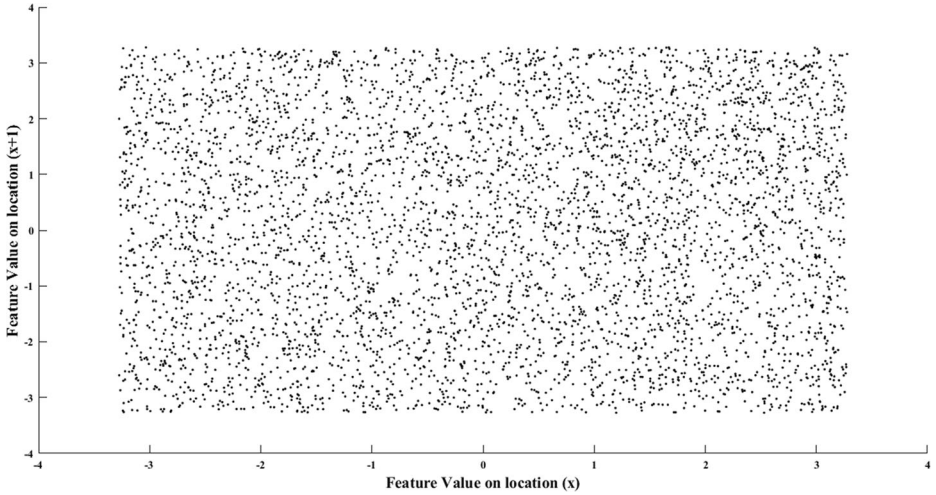


Fig. 12 Correlation plot of Encrypted Feature set

5.1.5 Histogram test

Histogram is a graph that shows the difference between feature values in a feature set. The histogram is drawn between the values of feature and the number of times each feature value occurrences [20]. The histogram is uniform if the value of encrypted features is uniformly distributed. Figure 13a-c depicts the histogram of the original, encrypted and decrypted feature sets. The encrypted feature set is uniform in Fig. 13b, and the proposed encryption scheme resist statistical attack.

5.1.6 Key space analysis

When the key space is limited, the comprehensive search breaks the cryptosystem, resulting in a brute force attack [20]. If the key space is larger than 2^{30}, the encryption system can resist this attack. In our proposed cryptosystem, the initial values and control parameters of the 3D logistic map(x_0, y_0, z, α, β, γ) are used as a secret key. The precision of one key is 10^{-12}, then key space is $10^{12} \times 10^{12} \times 10^{12} \times 10^{12} \times 10^{12} \times 10^{12} = 10^{72}$, which is equivalent to 2^{239}. The key space of the proposed system is greater than 2^{30}, so it resists brute force attacks.

5.2 Analysis the proposed System's performance

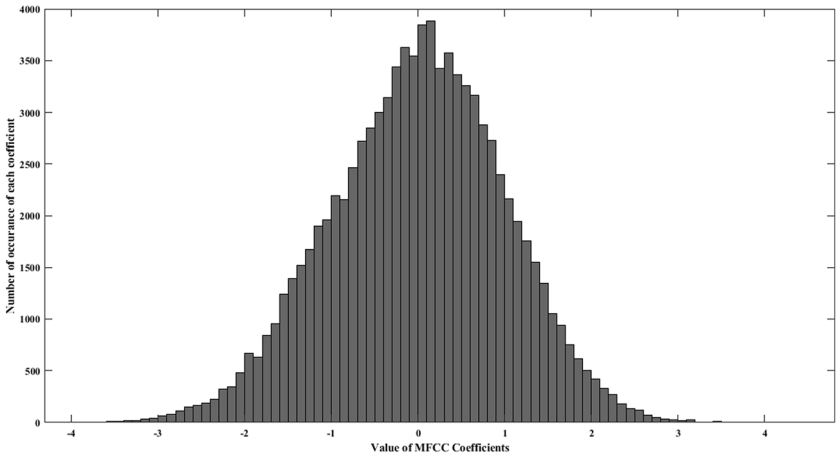
The free speech phrase and reading phrase of AVSpoof database utterances are concatenated during the training phase for each speaker. Training models for 44 speakers are generated using MFCC features derived from the training utterance set. In the text-dependent authentication method, the reading section of the AVSpoof database is used as a test speech. Before being sent via an insecure channel, the MFCC features retrieved from test speech are encrypted. The Performance of the text dependent authentication system is calculated by applying test features and decrypted test features on templates of all enrolled speakers. Two hundred segments are taken from decrypted/original features for testing. The matching is done between each test feature set and trained models. Various performance analysis measures such as ROC curves, accuracy and confusion matrix are used to analyse the efficiency of the identification system.

The confusion matrix was tabulated for computing each test speaker's individual recognition accuracy. This matrix is a square matrix in which the predicted class is represented by the columns and the experimental class is represented by the rows. The correctly categorised samples are represented by the diagonal of this matrix, whereas off-diagonal samples show the classification error. The text-dependent system's confusion matrix is shown in Table 4. Table 4 illustrates that the proposed system properly identified 199 of the 200 test segments of speaker 8 as speaker 8, and one as speaker 19. This means the person has 99.5% recognition accuracy.

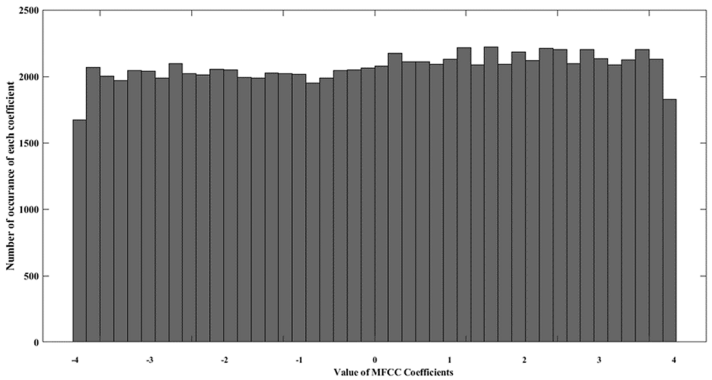
Recognition accuracy is used to determine the performance of the person identification systems [34]. The recognition accuracy is the ratio between the number of times the original test feature/decrypted test feature segments are correctly recognized to the total number of test original test feature/decrypted test feature. The recognition accuracy is expressed as (39)

$$\text{Accuracy} = \frac{\text{Correctly identified original test feature/decrypted test feature segments}}{\text{Total number of original test feature/decrypted test feature speech segments}} \tag{39}$$

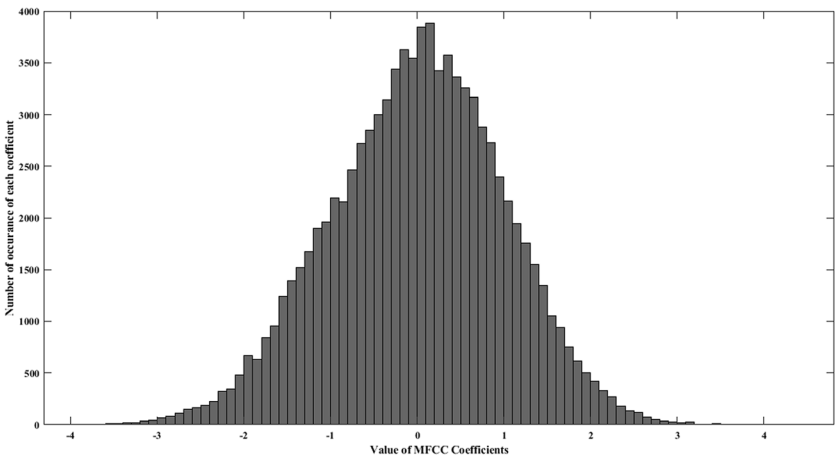
Tables 5 and 6 include the individual recognition accuracy of individual speakers for text independent and text dependent authentication system. Figure 14 depicts the individual



(a)



(b)



(c)

Fig. 13 **a** Histogram -Original Feature set. **b** Histogram - Encrypted Feature set. **c** Histogram - Decrypted Feature set

Table 4 Confusion matrix- Text dependent System

	Sp1	Sp2	Sp3	Sp4	Sp5	Sp6	Sp7	Sp8	Sp9	Sp10	Sp11	Sp12	Sp13	Sp14	Sp15	Sp16	Sp17	Sp18	Sp19	Sp20	Sp21	Sp22	Sp23	Sp24	Sp25	Sp26	Sp27	Sp28	Sp29	Sp30	Sp31	Sp32	Sp33	Sp34	Sp35	Sp36	Sp37	Sp38	Sp39	Sp40	Sp41	Sp42	Sp43	Sp44
Sp1	161	0	1	0	12	0	2	0	0	0	0	0	0	11	0	0	0	0	0	0	0	0	0	3	0	2	3	0	2	0	0	0	0	0	0	0	1	0	0	2	0	0
Sp2	0	194	0	0	0	0	0	1	1	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp3	0	0	173	0	0	0	6	2	2	0	0	0	0	0	0	1	0	0	0	0	5	0	2	0	0	0	1	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp4	0	0	0	192	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	5	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1
Sp5	0	0	0	0	195	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp6	0	0	1	0	0	182	1	2	0	0	0	0	0	0	4	0	0	4	0	0	0	3	6	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp7	0	0	0	0	0	0	198	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
Sp8	0	0	0	0	0	0	0	199	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp9	0	0	0	0	0	0	0	0	199	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp10	0	0	1	0	0	0	0	1	1	179	0	0	0	0	0	11	0	0	0	0	0	0	4	0	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp11	0	0	0	0	0	0	4	0	0	193	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp12	0	0	0	0	0	1	0	0	0	197	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp13	0	0	0	0	2	0	1	4	0	0	0	188	0	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp14	0	0	0	0	1	0	0	0	0	0	0	0	199	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp15	0	0	1	0	0	0	1	9	1	0	0	1	1	0	159	1	0	0	5	0	0	0	19	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp16	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	199	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp17	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	200	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp18	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	2	195	0	0	0	1	0	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp19	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	198	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp20	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	193	0	0	1	0	1	0	0	1	0	0	1	0	0	0	0	0	0	0	1	0	2	0	0
Sp21	0	0	0	1	0	0	1	0	1	0	0	0	1	0	0	4	0	2	0	189	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp22	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	200	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp23	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	197	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp24	0	0	0	0	0	6	1	0	0	0	0	0	0	1	9	0	1	0	0	0	0	0	181	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp25	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	198	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp26	0	0	0	0	0	0	0	0	1	0	2	0	0	0	0	0	0	0	1	0	0	0	2	0	194	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp27	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	10	0	0	0	0	0	0	0	0	0	188	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp28	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	2	0	0	0	0	0	0	0	0	0	0	195	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp29	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	198	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	199	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp31	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	198	0	0	0	0	0	0	0	0	0	0	0	0
Sp32	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	192	1	0	0	2	0	0	5	0	0	0	0
Sp33	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	196	0	1	0	0	0	1	0	1	1	0
Sp34	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	36	0	162	0	0	0	0	0	0	0	2	0
Sp35	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	195	0	1	0	3	1	0	0	0
Sp36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	198	0	0	1	0	1	0	0
Sp37	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	2	1	172	1	0	15	0	6	0	1
Sp38	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	197	0	1	0	1	1	0
Sp39	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	2	196	1	0	0	0
Sp40	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	200	0	0	0
Sp41	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	199	0	0	0
Sp42	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	11	0	123	66	0
Sp43	0	0	2	0	0	31	2	0	0	0	0	0	16	0	0	0	0	0	0	0	10	0	0	0	0	0	0	0	0	0	13	0	20	0	4	0	0	0	0	0	102	0
Sp44	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1	5	0	12	0	4	1	1	0	1	1	4	2	165

accuracy of each speaker for text dependent authentication system. Text dependent identification system's overall recognition accuracy is 96%. In text independent authentication system, pass portion speeches of AVSpoof database are taken as test speech. Figure 15 depicts the recognition accuracy of text-independent system for individual speaker. Tables 5 and 6 show that the text independent and text dependent authentication method has an overall recognition rate of 96% and 88% respectively. Figure 16 indicates the performance of the system for various attacks.

Table 5 Confusion matrix- Text independent System

	Sp1	Sp2	Sp3	Sp4	Sp5	Sp6	Sp7	Sp8	Sp9	Sp10	Sp11	Sp12	Sp13	Sp14	Sp15	Sp16	Sp17	Sp18	Sp19	Sp20	Sp21	Sp22	Sp23	Sp24	Sp25	Sp26	Sp27	Sp28	Sp29	Sp30	Sp31	Sp32	Sp33	Sp34	Sp35	Sp36	Sp37	Sp38	Sp39	Sp40	Sp41	Sp42	Sp43	Sp44
Sp1	130	0	0	6	0	0	0	0	24	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	28	11	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp2	3	138	3	3	8	1	6	0	0	6	0	0	0	0	0	1	0	4	2	0	0	0	0	0	12	6	5	1	2	0	1	0	0	0	0	7	0	0	0	0	1	2
Sp3	5	2	125	11	0	0	0	7	1	0	0	0	0	1	5	7	0	0	1	1	0	0	0	6	0	13	3	0	0	0	2	0	0	1	0	0	0	0	0	0	0	0
Sp4	0	0	0	144	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	56	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp5	39	0	0	0	161	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp6	11	0	0	0	4	92	0	0	0	0	0	0	0	1	0	0	0	0	0	0	2	0	1	35	2	0	0	0	0	0	52	0	0	0	0	0	0	0	0	0	0	0
Sp7	4	0	3	3	0	0	114	0	0	10	0	0	12	4	17	0	3	0	2	8	0	0	0	3	4	3	6	0	0	2	7	0	0	0	0	1	0	0	0	0	0	0
Sp8	0	0	0	0	0	0	0	194	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp9	0	0	0	0	0	0	0	0	166	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	28	0	3	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp10	0	0	0	0	0	0	0	0	200	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp11	1	1	0	0	0	5	0	0	0	102	0	0	1	9	1	0	1	12	7	4	2	0	1	0	12	1	2	1	0	0	0	0	0	4	0	0	0	0	0	11	0	0
Sp12	1	0	4	0	1	4	0	5	0	7	0	41	0	6	0	6	0	11	5	0	26	0	1	0	0	59	6	0	1	7	0	0	3	0	5	0	0	7	0	0	0	0
Sp13	0	0	0	1	4	0	9	0	0	0	5	0	105	0	10	0	0	0	0	4	0	0	0	5	7	25	6	2	0	0	0	0	0	0	0	1	2	0	0	19	1	0	0
Sp14	0	0	0	0	0	0	0	7	0	0	0	0	0	157	2	0	0	0	0	24	0	10	0	0	0	0	6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp15	0	0	0	0	0	0	0	24	0	31	0	0	0	0	140	0	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp16	0	13	0	0	0	0	0	0	0	0	0	41	0	0	0	137	0	0	0	0	0	0	0	0	0	0	6	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp17	0	0	0	0	0	0	0	0	15	0	0	0	4	0	172	0	0	0	0	0	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp18	10	0	2	19	5	19	1	0	0	0	1	12	4	2	103	6	6	0	2	0	0	0	1	6	0	0	0	0	0	0	6	0	0	0	0	0	0	0	1	0	0	0
Sp19	0	0	0	0	1	0	23	0	0	0	0	0	0	0	0	156	0	0	0	0	0	0	0	0	0	20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp20	4	0	3	12	0	0	3	0	2	0	0	3	0	0	1	0	116	1	1	0	4	7	12	0	0	1	0	0	0	0	11	0	2	1	0	0	16	0	0	0	0	0
Sp21	0	0	0	0	67	0	0	0	0	0	0	0	0	0	0	0	0	36	97	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp22	20	2	2	5	2	2	1	6	11	10	0	0	0	5	3	0	4	0	0	3	2	80	0	3	4	0	15	0	6	0	1	0	0	0	0	10	0	0	0	0	0	3
Sp23	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	200	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	149	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp25	0	0	19	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	176	5	6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp26	14	0	6	35	0	0	0	5	0	0	0	0	0	4	7	0	1	8	0	0	1	0	0	1	0	49	10	14	2	8	0	4	0	0	0	6	7	0	0	0	1	0
Sp27	0	0	29	0	0	0	0	0	0	0	0	0	0	0	0	17	0	0	0	0	0	0	0	0	0	0	154	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp28	0	1	0	8	13	0	0	0	0	9	3	0	0	0	7	0	4	1	0	0	2	0	0	0	2	0	16	126	1	3	2	0	0	0	0	0	0	0	0	2	0	0
Sp29	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	70	0	0	0	130	0	0	0	0	0	0	0	0	0	0	0	0
Sp30	0	0	7	4	6	1	0	0	0	0	0	0	0	0	0	10	0	0	0	0	0	4	0	0	1	1	15	0	123	0	7	0	1	7	0	7	0	5	0	0	1	0
Sp31	0	1	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	55	0	0	0	0	0	1	0	0	131	0	0	0	0	0	0	10	0	0	0	0	0
Sp32	2	0	0	16	1	0	2	0	0	1	0	0	0	0	7	0	5	0	0	0	0	2	0	0	2	6	4	0	1	1	69	6	26	3	0	16	9	1	2	0	2	19
Sp33	0	0	0	0	0	1	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	190	1	0	0	0	0	0	0	0	0	0
Sp34	0	0	0	0	1	0	0	0	0	0	0	0	0	4	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	19	2	128	8	0	13	5	4	2	0	0	7
Sp35	0	1	0	7	1	0	0	0	0	2	0	0	0	0	0	2	0	6	0	1	0	2	1	2	6	0	0	0	0	0	3	0	98	0	13	3	13	4	8	25	5	3
Sp36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	200	0	0	0	0	0	0	0	0
Sp37	2	0	1	12	0	4	0	0	8	0	0	0	0	4	0	0	0	0	12	1	0	0	11	1	0	0	0	0	0	0	13	5	19	5	6	76	9	0	7	0	2	2
Sp38	0	0	0	27	0	0	17	0	0	0	0	0	17	0	0	0	0	0	0	27	0	0	0	0	0	0	0	0	0	0	0	2	0	18	0	16	50	21	0	0	5	0
Sp39	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	26	0	0	0	0	0	0	170	0	0	0	0
Sp40	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	29	22	31	0	0	0	0	0	96	0	0	22
Sp41	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	165	35	0
Sp42	0	0	0

Table 6 Performance analysis of the proposed System tested with Genuine and Attack utterances

Speaker	Genuine Test Utterances		Attack Test Utterances				
	Text dependent system Recognition Accuracy	Text independent system Recognition Accuracy	Rejection Rate				
			Voice conversion Physical access Attack	Speech synthesis Attack	Replay laptop HQ speaker Attack	Replay laptop Attack	Replay phone Attack
Speaker1	80.50	93.65	51.96	89.00	62.75	61.00	89.00
Speaker2	97.00	96.97	100.00	100.00	100.00	100.00	100.00
Speaker3	86.50	94.37	100.00	100.00	100.00	100.00	100.00
Speaker4	96.00	62.03	100.00	99.00	98.04	100.00	99.00
Speaker5	97.50	68.67	64.71	98.00	100.00	100.00	98.00
Speaker6	91.00	74.32	99.02	100.00	100.00	100.00	100.00
Speaker7	99.00	100.00	100.00	100.00	100.00	100.00	100.00
Speaker8	99.50	91.94	100.00	100.00	100.00	100.00	100.00
Speaker9	99.50	98.46	100.00	98.00	100.00	100.00	98.00
Speaker10	89.50	90.77	99.02	89.00	52.94	97.00	89.00
Speaker11	96.50	100.00	100.00	100.00	100.00	100.00	100.00
Speaker12	98.50	91.43	100.00	100.00	100.00	100.00	100.00
Speaker13	94.00	98.28	100.00	100.00	100.00	100.00	100.00
Speaker14	99.50	93.18	100.00	98.00	100.00	100.00	98.00
Speaker15	79.50	82.35	96.08	99.00	99.02	99.00	99.00
Speaker16	99.50	98.65	98.04	100.00	100.00	100.00	100.00
Speaker17	100.00	95.24	98.04	99.00	100.00	100.00	99.00
Speaker18	97.50	89.23	100.00	100.00	100.00	100.00	100.00
Speaker19	99.00	96.30	100.00	100.00	100.00	100.00	100.00
Speaker20	96.50	95.59	100.00	100.00	100.00	100.00	100.00
Speaker21	94.50	84.75	97.06	100.00	99.02	100.00	100.00
Speaker22	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Speaker23	98.50	98.46	100.00	100.00	99.02	94.00	100.00
Speaker24	90.50	96.61	100.00	100.00	100.00	100.00	100.00
Speaker25	99.00	98.33	100.00	100.00	100.00	100.00	100.00
Speaker26	97.00	38.89	99.02	100.00	99.02	100.00	100.00
Speaker27	94.00	96.30	100.00	96.00	93.14	100.00	96.00
Speaker28	97.50	97.01	100.00	100.00	100.00	100.00	100.00
Speaker29	99.00	82.09	100.00	100.00	100.00	100.00	100.00
Speaker30	99.50	92.65	100.00	100.00	100.00	100.00	100.00
Speaker31	99.00	91.18	95.10	99.00	100.00	100.00	99.00
Speaker32	96.00	81.25	100.00	100.00	100.00	100.00	100.00
Speaker33	98.00	87.01	95.10	96.00	100.00	100.00	96.00
Speaker34	81.00	80.56	100.00	100.00	100.00	100.00	100.00
Speaker35	97.50	100.00	100.00	97.00	100.00	100.00	97.00
Speaker36	99.00	98.65	100.00	100.00	97.06	100.00	100.00
Speaker37	86.00	55.41	100.00	100.00	96.08	100.00	100.00
Speaker38	98.50	81.82	100.00	98.00	99.02	100.00	98.00
Speaker39	98.00	94.74	89.22	93.00	87.25	100.00	93.00
Speaker40	100.00	98.41	98.04	96.00	100.00	100.00	96.00
Speaker41	99.50	84.62	100.00	100.00	100.00	100.00	100.00
Speaker42	100.00	78.26	98.04	75.00	93.14	98.00	75.00
Speaker43	97.00	97.40	86.27	86.00	96.08	100.00	86.00
Speaker44	95.50	69.64	100.00	99.00	95.10	100.00	99.00

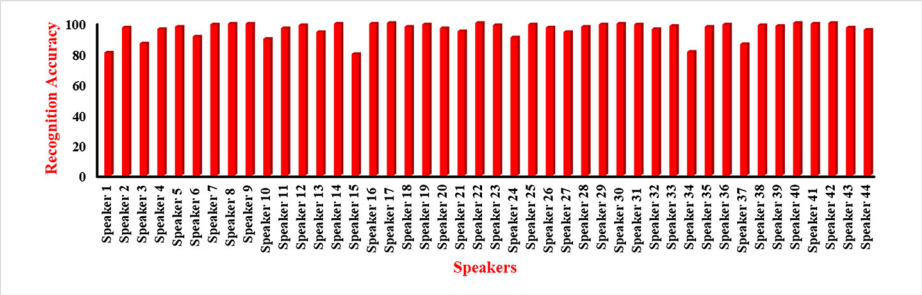


Fig. 14 Individual Accuracy of text dependent authentication system

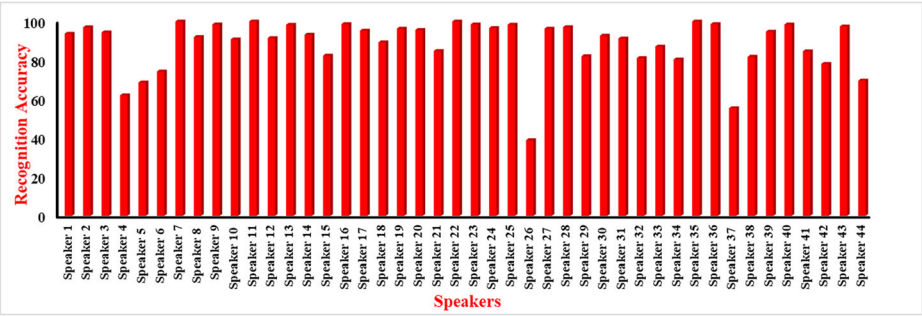


Fig. 15 Individual Accuracy of textindependent authentication system

5.2.1 Proposed system's performance with attack utterances

The speech-based person authentication system is affected by three types of spoofing attacks: voice conversion, replay and speech synthesis attack. A replay attack is a replay of a pre-recorded speech to an authentication system. Voice conversion is a method that changes attack speaker's speech to the target speaker's speech. Text to speech synthesis is the artificial production of speech from normal language text. These attack utterances are taken from

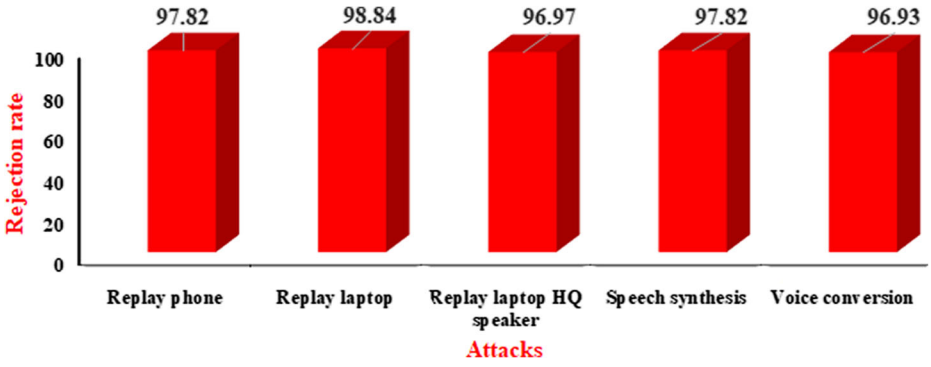


Fig. 16 Performance of proposed system tested with attack utterances

AVSpoof database. The proposed authentication system analysed in terms of true rejection rate tested with attack utterances. The percentage of times the system accurately rejects a false claim of identity is called true rejection rate. Table 6 includes the individual rejection rate of the proposed authentication system tested with different sets of attack utterances.

5.2.2 Receiver operating characteristics (ROC) curve

The ROC curves are used to analyse the efficiency of the identification systems. ROC graph is plotted among the false positive rate and true positive rate [3]. Intersecting points of false positive rate and true positive rate lying above the boundary line reveals the efficiency of the person identification system for genuine and attack test utterances. Figure 17 shows the ROC curve of text dependent system being tested with genuine utterance. From Fig. 17, it can be perceived that the feature selection and choice of machine learning technique would prove to be better in getting high acceptance rate. Figure 18 depicts the ROC curve of the proposed system, tested with replay phone attacks utterance. From Figure 18, it is apparent that the ROC curves are grouped at the bottom right corner, which indicates that this system has high rejection rate against attacks utterance.

5.3 Comparison of the proposed system with conventional speech based crypto system

Many researchers have worked on developing encryption techniques based on chaotic maps. Comparative analysis with respect to the Methods used in encryption, security analysis techniques used in the proposed and conventional systems are depicted in Table 7. The performance of the security algorithm and person authentication system of the proposed and conventional systems is listed in Table 8. The correlation coefficient computed between encrypted and original signal of the proposed system is lesser than the conventional systems [21, 29], indicating that the encrypted signal is random and uncorrelated with the original signal. The value of SNR and PSNR between the original and encrypted signal of the proposed

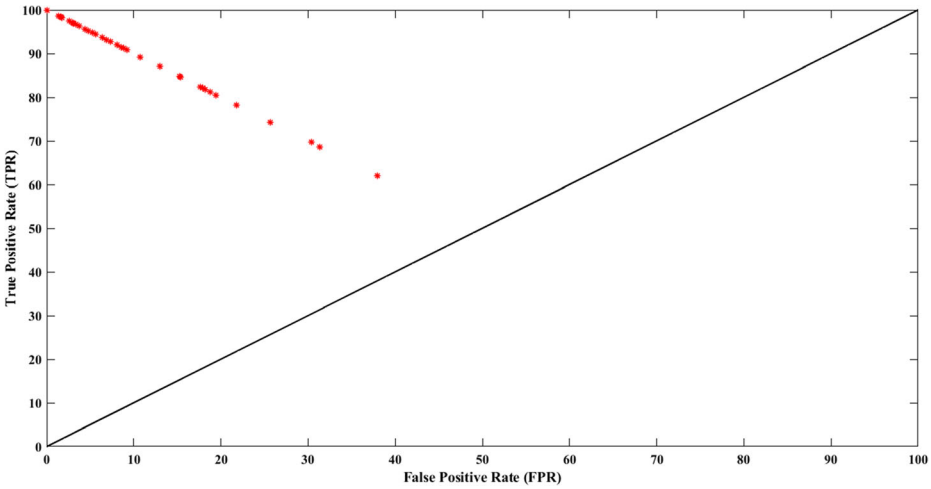


Fig. 17 ROC Curve of textdependent authentication system tested with genuine utterances

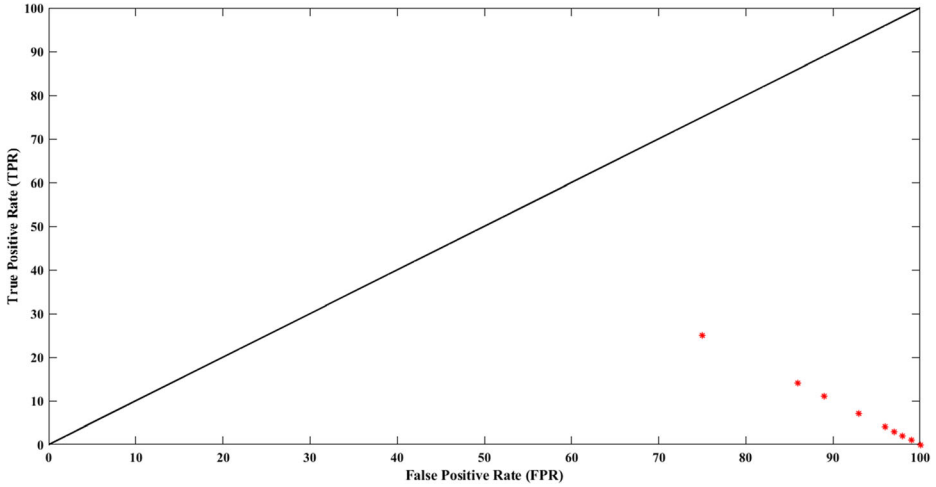


Fig. 18 ROC Curve of proposed system tested with Replay phone attacks utterance

system is very less compared to the existing systems [21, 29], it shows that the encrypted signals are highly undetectable. Table 9 depicts the comparison of the recent person authentication systems implemented using different biometrics.

5.4 Transmission bandwidth of the proposed system

In AVSpoof database, Speeches are captured at a 44.1KHz sampling frequency and downsampled to 16 KHz and every sample is represented with 16 bits. In traditional systems, the test speech samples are encrypted and transmitted to the authentication system.

In this work, test features extracted from test speech set are encrypted and transmitted to the authentication system. The test speeches are divided into little segments. The duration of the segment is 25 ms (400 samples) and frame shift is 10 ms (160 samples). Each feature is

Table 7 Comparative analysis of the Proposed System with Existing Speech based Crypto systems

Speech Based Encryption System	Methods used for Encryption	Security Analysis Matrics Used	Input to the Encrypted System
R. Nagakrishnan., et.al [21]	Choas based – Henon, Sine, Tent, Logistic Maps and DNA encoding	PSNR, SNR, Correlation, Histogram, Keyspace, NSCR, UACI	Speech Samples
Farsana F J., et.al [9]	Choas based – Duffing Map	Keyspace, Correlation, SNR	Speech Samples
W. S. Sayed., et.al [30]	Choas based – Modified Logistic and tent Map	PSNR, SNR, Correlation, Histogram,	Speech Samples
Sheela S. J., et.al [32]	Choas based - Modified Henon map	Correlation, Key sensitivity, Histogram, SNR, PSNR	Speech Samples
Sathiyamurthi. P., et.al [29]	hybrid-hyper chaotic system	Histogram, Spectrogram, Correlation, SNR, PSNR, Key sensitivity	Speech Samples
Proposed System	Choas based – 3D Logistic Maps and DNA encoding	PSNR, SNR, Correlation, Histogram, Key space, NFCR, UACI	Features extracted from speech

Table 8 Comparative analysis based on the metrics used - proposed and conventional system

	Method	Proposed system	R.Nagakrishnan., et.al [21]	R.Nagakrishnan., et.al [22]	Sathiyamurthi. P., et.al [29]
Security Analysis of Speech encryption System1.11.1	Correlation coefficient	0.0027	0.55	–	0.9875
	PSNR	−91.55	5.672	–	50.21
	SNR	−91.55	−13.62	–	123.57
	NSCR	99.9977	99.9999	–	99.981
	UACI	36.56	49.99	–	33.059
Performance Analysis of Person Authentication System	Features Used	MFCC	MFCC	PLPC	–
	Modeling Techniques	RNN	K-Means	K-Means	–
	Accuracy of Text independent System	88%	66%	81%	–
	Accuracy of Text dependent System	96%	97%	96%	–

Table 9 Person authentication system using different biometrics

S. No	Person Authentication system	Biometric Used
1	Das, B.B., et.al [2]	Electroencephalography (EEG) signal
2	Sathiya Lakshmanan., et.al [15]	Finger knuckle print
3	Juhong, A., et.al [13]	Face
4	Resmi, P., et.al [25]	Face
5	Proposed system	Speech

represented by 16 bits. Comparative analysis with respect to the bandwidth requirement of the proposed and conventional system is depicted in Table 10. Result shows that the transmission bandwidth requirement of the proposed system is 10 times lesser than the conventional system.

5.5 Time analysis

Time complexity measurement for performing the proposed cryptosystem is done and compared with the conventional system [21]. Time is taken to encrypt the features derived from the speech and speech is listed in Table 10. Result shows that the proposed features

Table 10 Bandwidth requirements and Time complexity analysis of the Proposed and Conventional Authentication systems based on Speech frame

Parameters	Conventional System [21] (Speech based)	Proposed System (Feature based)
Input to the Encryption System	Speech	Feature
Number of samples or Coefficients in 25 ms speech frame	400	13
Required channel bandwidth	6400 bits per 25 ms frame	208 bits per 25 ms frame
Time taken for one frame	100 msec per 25 ms frame	0.00325 msec per 25 ms frame

based crypto system requires 0.00325 to encrypt one speech frame. The proposed feature based encryption system takes less time as compared to the conventional speech based encryption system.

6 Conclusion

Feature based secured Speech based authentication system is proposed. The MFCC features are extracted from training speech set and recurrent neural network is trained with features and a template is created for each person. The Crypto system is developed to secure the MFCC features extracted from test speech set. The proposed encryption system is based on 3D chaotic map and DNA Encoding. The initial values and control variables of 3D logistic map are generated from the hash value, which is produced by SHA 3 algorithm. Two-level diffusion stages and confusion stage gives an encrypted feature set. The encryption system's performance is evaluated using several metrics based on statistical tests. The results show that the crypto system is resistant to statistical, differential and brute force assaults. According to the results, the encryption scheme is opposed to statistical, differential and brute force attacks. The proposed authentication system is evaluated against genuine and spoofing attacks utterances. Performance metrics such as recognition accuracy and ROC curve are used to analyze the performance of the proposed system. According to these results, it is revealed that the authentication using genuine utterances has provided the identification rate of 88% for text independent system and 99% for text dependent system, whereas there has been a high rejection for authentication system tested with attacks utterance. Our proposed system helps remote access of database, online transactions and finds applications in defence sectors, banking sectors and so on using inherently present speech as a biometric for authenticating persons. Feature based security system provides minimization in transmission bandwidth requirement, time complexity and computational complexity in implementation of cryptosystem as compared to the speech signal based cryptosystem for security. In addition, feature extraction is not needed to be done in the receiving system for using feature based crypto system for security.

Code availability The code that supports the findings of this study is available from the corresponding authors on reasonable request.

Authors' contributions R.Nagakrishnan and A. Revathi contributed to the design and implementation of the research, to the analysis of the results and to the writing of the manuscript.

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Data availability All relevant data are within the article and its supporting information files

Declarations

Conflict of interest The authors have declared that there is no conflict of interest exists.

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